

Paleointensity

Recovering ancient field strength from imperfect recorders

Thermoremanence • experiments • failure modes • inference • modern workflows

CENTRAL IDEA

A palaeointensity is not a number read from a rock; it is an inference that must remain credible after physical, experimental, and statistical tests.

Paleointensity asks an inverse question

Given a remanence measured today, what field strength was present when the record formed?



Each part of this has assumptions about recording, preservation, and laboratory equivalence.

By the end, you should be able to defend a field estimate

- Connect TRM physics to the slope measured in a Thellier-style experiment.
- Recognize common failure modes before a “good-looking” Arai plot becomes a result.
- Match a protocol to material, experiment budget, and the likely source of bias.
- Report uncertainty and selection choices as part of the scientific result.

A WORKING STANDARD

The best result is the one whose interpretation is easy for another researcher to reproduce, access, and challenge.

Field intensity can inform core dynamics and Earth-system change

Geodynamo state

Long-lived intensity lows and highs constrain how the core generates and reverses its field.

Shielding and atmosphere

Field strength helps set the context for magnetospheric shielding and planetary habitability.

Chronology and correlation

Archaeomagnetic and volcanic field curves can add independent constraints to age models.

The intensity value matters only after its geological age, spatial representativeness, and uncertainty have been made explicit.

DEEPER DIVE

Look into the publications of John Tarduno, Richard Bono, Andy Biggin, Courtney Sprain, Yimeng Zhang, Julien Aubert, Cathy Constable, Monika Korte, Sanja Panovska...

Absolute and relative paleointensity solve different problems

ABSOLUTE PALAEOINTENSITY

Recover field strength in μT or virtual dipole moment by comparing an ancient remanence with a laboratory-induced remanence.

RELATIVE PALAEOINTENSITY

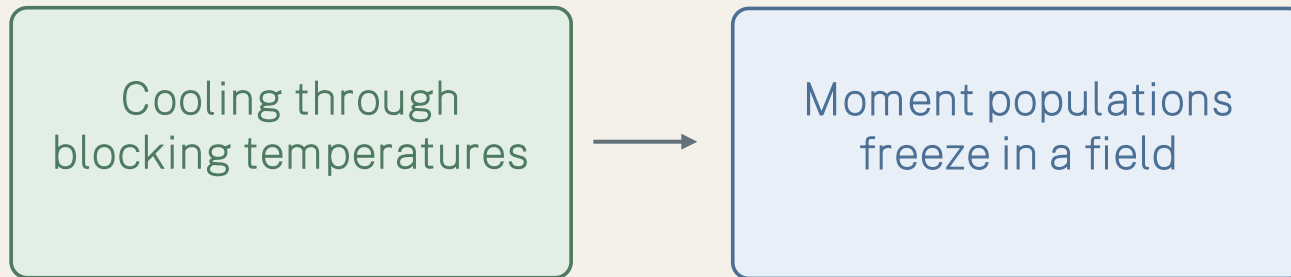
Track relative changes in sedimentary records after normalizing NRM by a magnetic concentration proxy (see Josh's lecture).

Complementary Records

Relative records can be superb time series, but their scale is not automatically an absolute field estimate. SINT2000, PADM2M are good examples of scaled records.

A rock records field strength through remanence

For a primary volcanic or archaeological record, the desired remanence is a thermal remanent magnetization (TRM).



All palaeointensity methods are based on:

$$\frac{NRM}{TRM_{lab}} = \frac{B_{ancient}}{B_{lab}}$$

The palaeointensity experiment asks whether a laboratory remanence can replace the natural record proportionally: I'll show that through sketch in a minute.

Ideal TRM is nearly proportional to weak applied field

In the linear, weak-field regime for an ideal assemblage:

$$M_{\text{TRM}} = C \cdot B$$

C contains the recorder's grain population, mineralogy, geometry, and cooling history.

We do not know C a priori. A palaeointensity protocol is an attempt to estimate it without changing it.

Sources: Tauxe Essentials; Selkin et al. (2007)

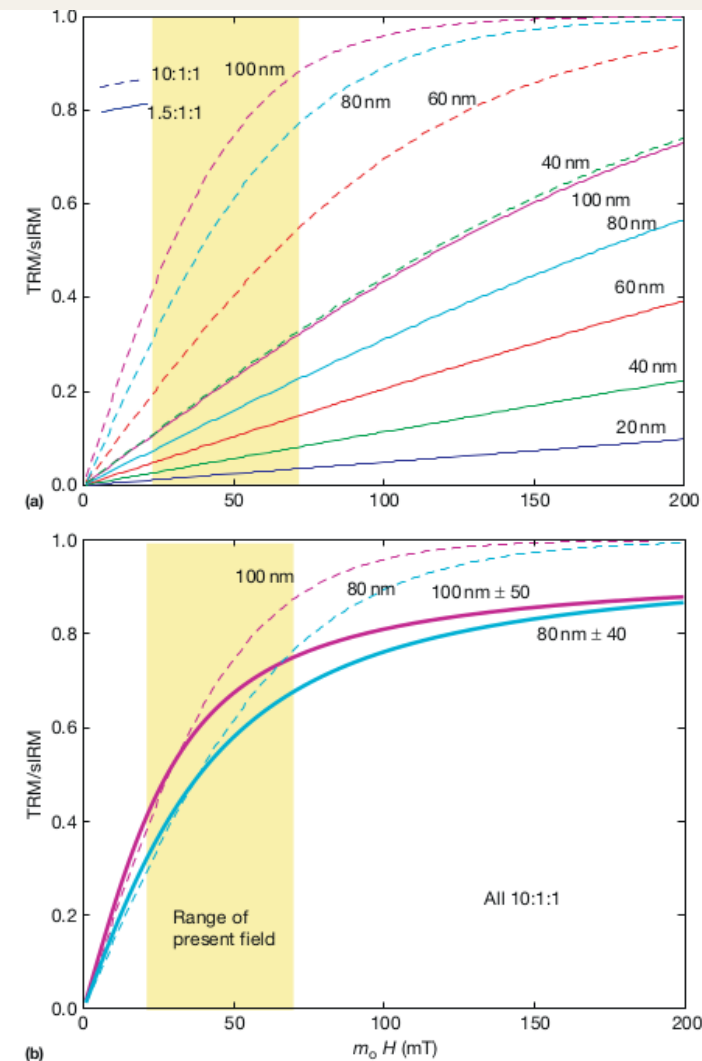
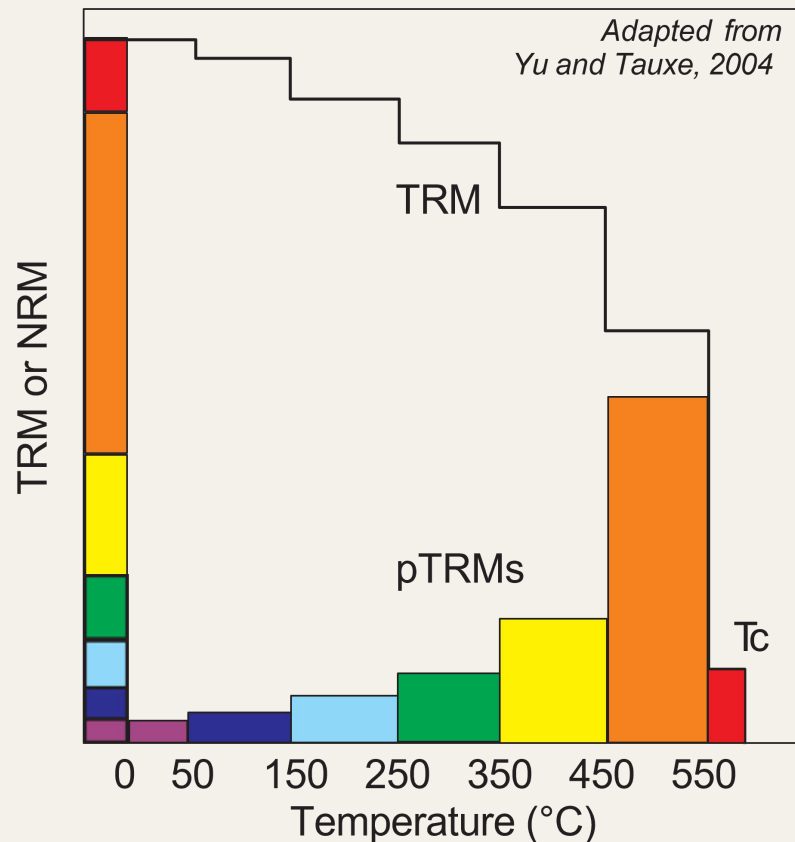


Figure 4 Predicted thermal remanent magnetization (TRM) expressed as a fraction of saturation for various particle sizes and distributions of magnetite. Note the nick point for which the linearity assumption fails is a strong function of particle size, but linearity holds true for equant particles in fields less than a hundred μT . Strongly elongate particles will behave in a more nonlinear fashion.

Blocking and unblocking link the ancient and laboratory remanences

Each grain population contributes over a range of temperatures, not at one perfectly sharp point, but we can divide TRM into partial TRM (pTRMs).



IMPLICATION

NRM pTRMs = Lab pTRMs

which means pTRMs are the same when cooling in nature and then reheating in the lab.

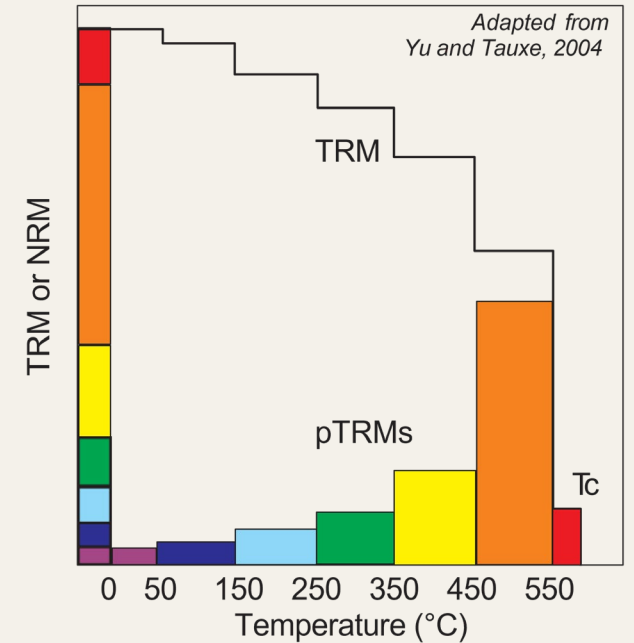
Blocking = unblocking

Thellier's laws

Reciprocity A pTRM unblocked at a temperature can be reacquired there.

Independence Distinct blocking-temperature fractions behave independently.

Additivity The total TRM is the sum of the relevant pTRMs.



A defensible estimate requires six tests

1. Primary remanence

The NRM interval represents the field of interest.

2. Stable recorder

Heating does not create or destroy the relevant carrier.

3. Comparable blocking populations

Laboratory pTRM and lost NRM share the same "unblockers".

4. Linear response

TRM scales with the applied field in the relevant range.

5. Magnetic anisotropy

Anisotropy is tested and corrected for.

6. Natural vs lab cooling rates

Cooling rate considered and corrected for.

01

THE EXPERIMENT

Replace magnetization without changing the recorder

The central difficulty is that thermal treatments can alter the object being measured.

The Thellier–Coe experiment

1

Demagnetize

Heat in zero field; measure lost NRM.

2

Impart pTRM

Reheat at the same T in a known laboratory field.

3

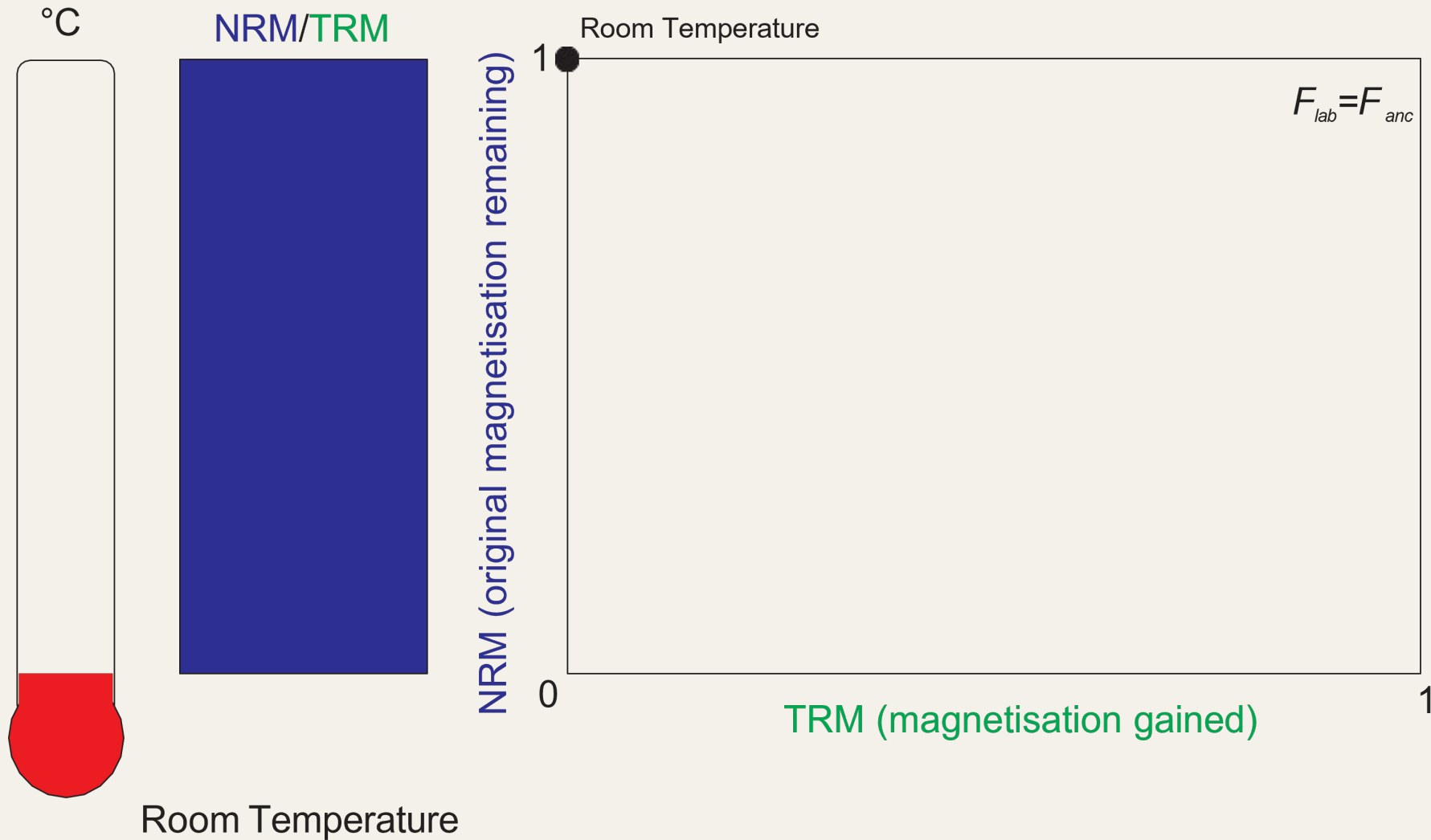
Compare

Relate NRM loss to laboratory pTRM gain.

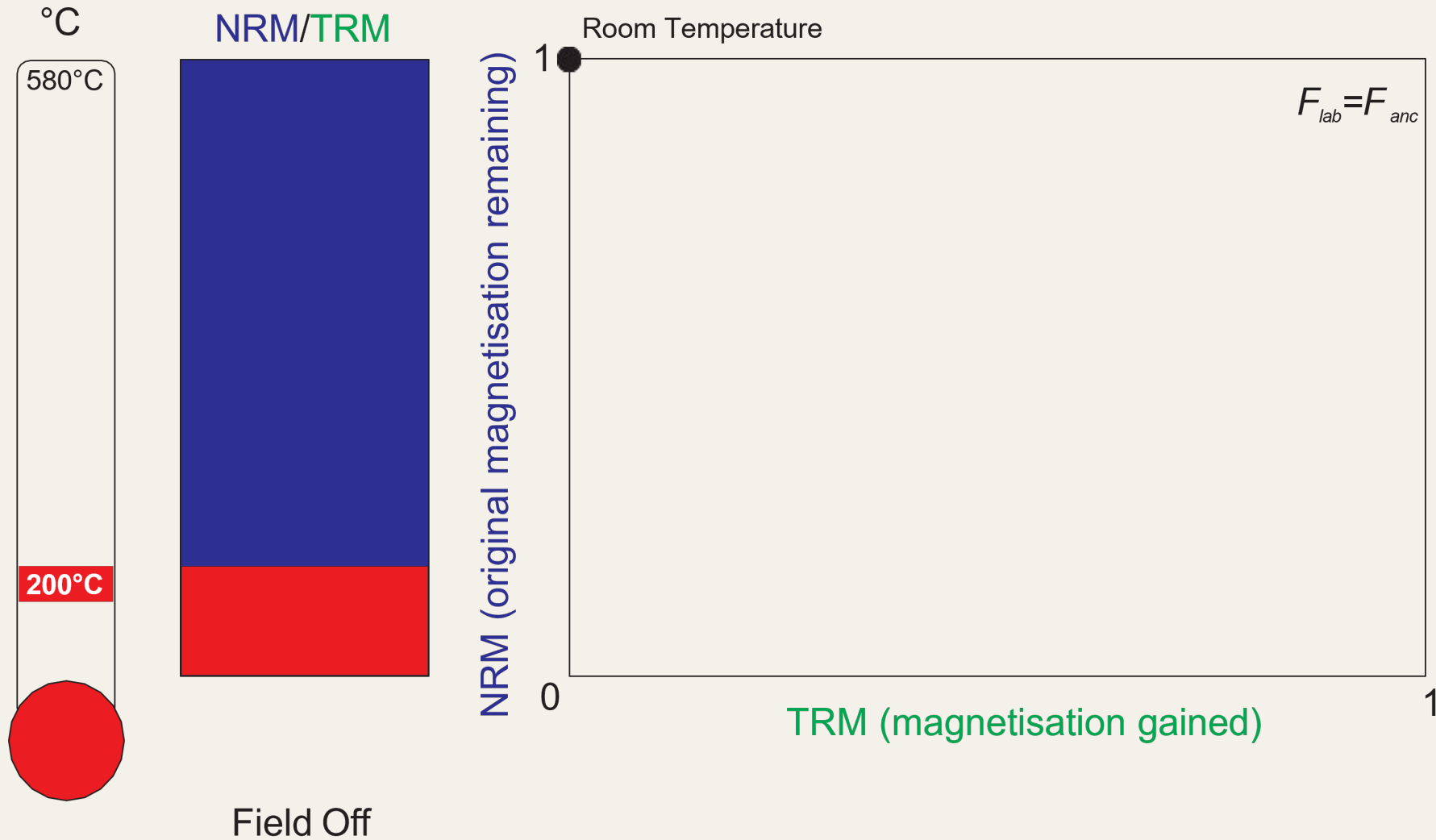
DESIGN PRINCIPLE

Use matched temperature steps so the same unblocking population is removed and replaced.

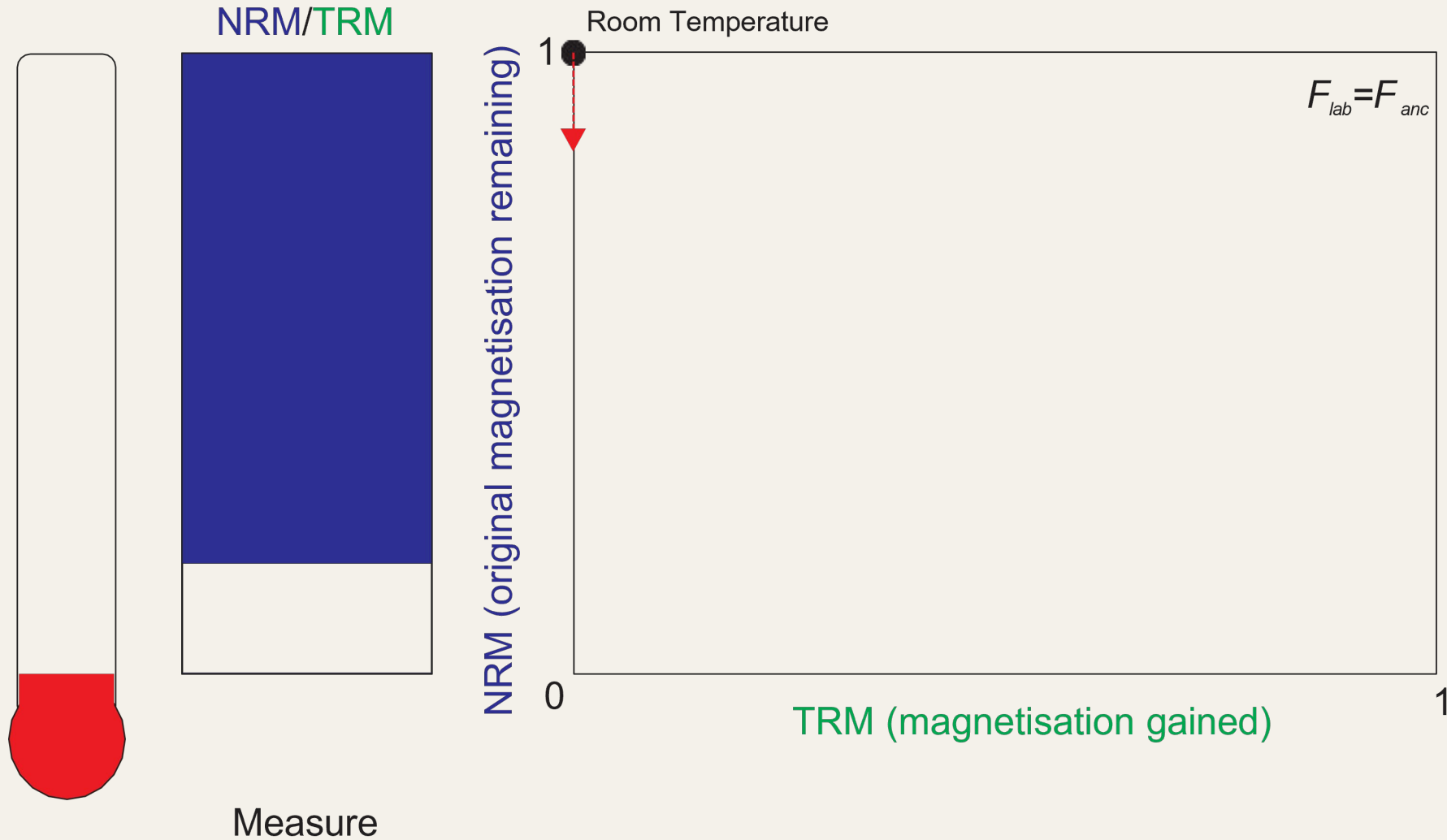
Coe-Thellier Method (on Arai-Nagata plot)



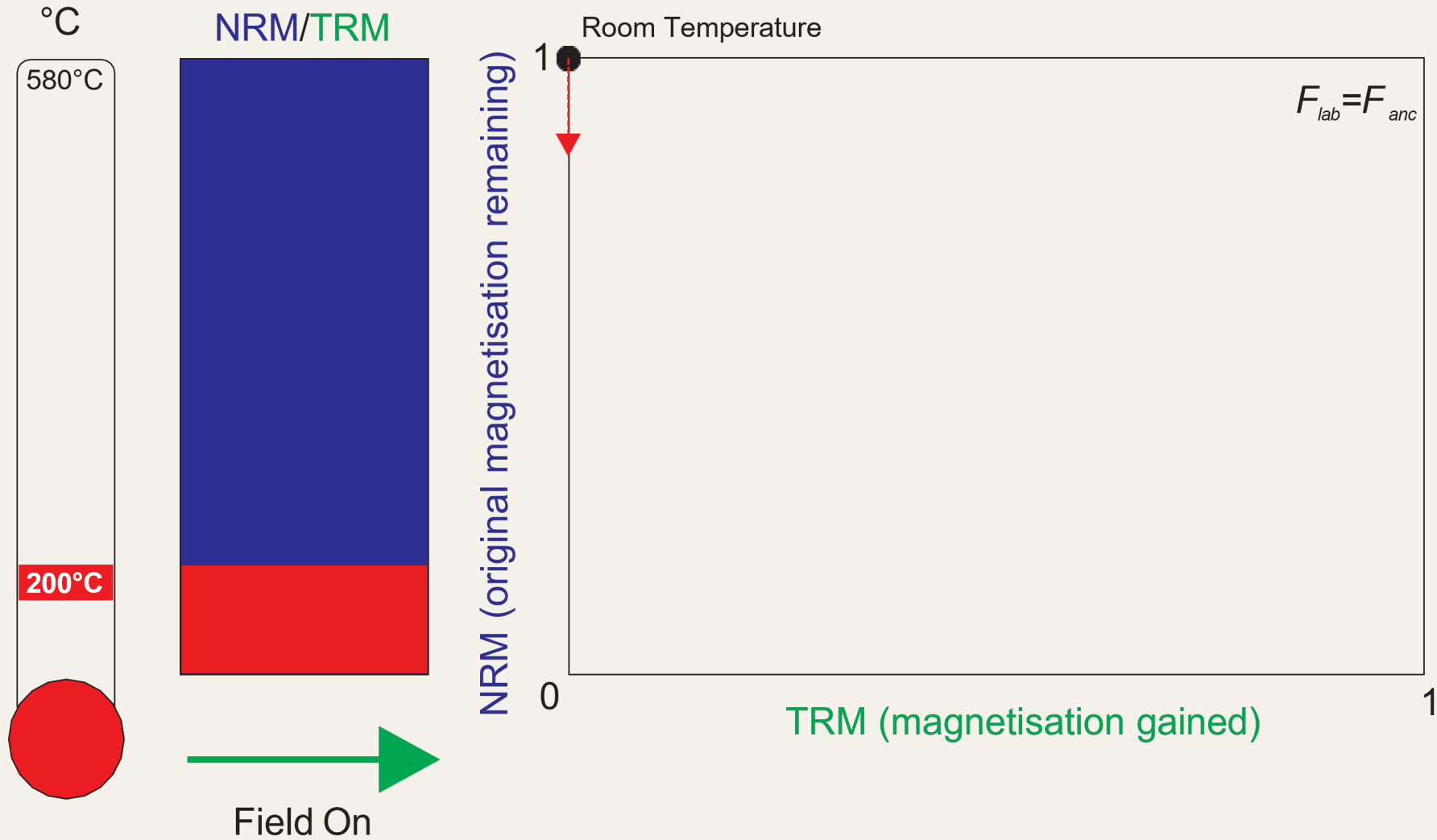
Coe-Thellier Method (on Arai-Nagata plot)



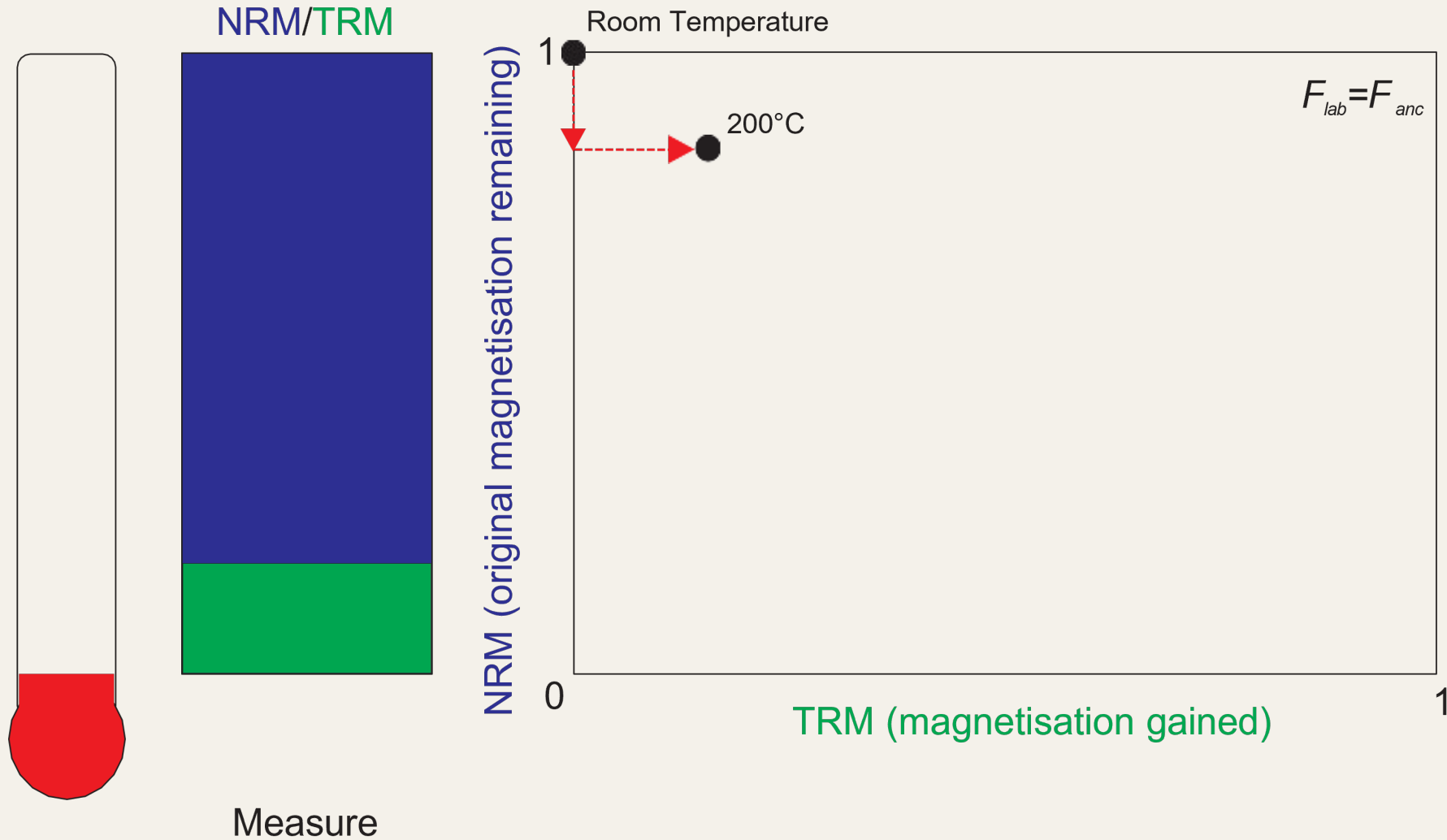
Coe-Thellier Method (on Arai-Nagata plot)



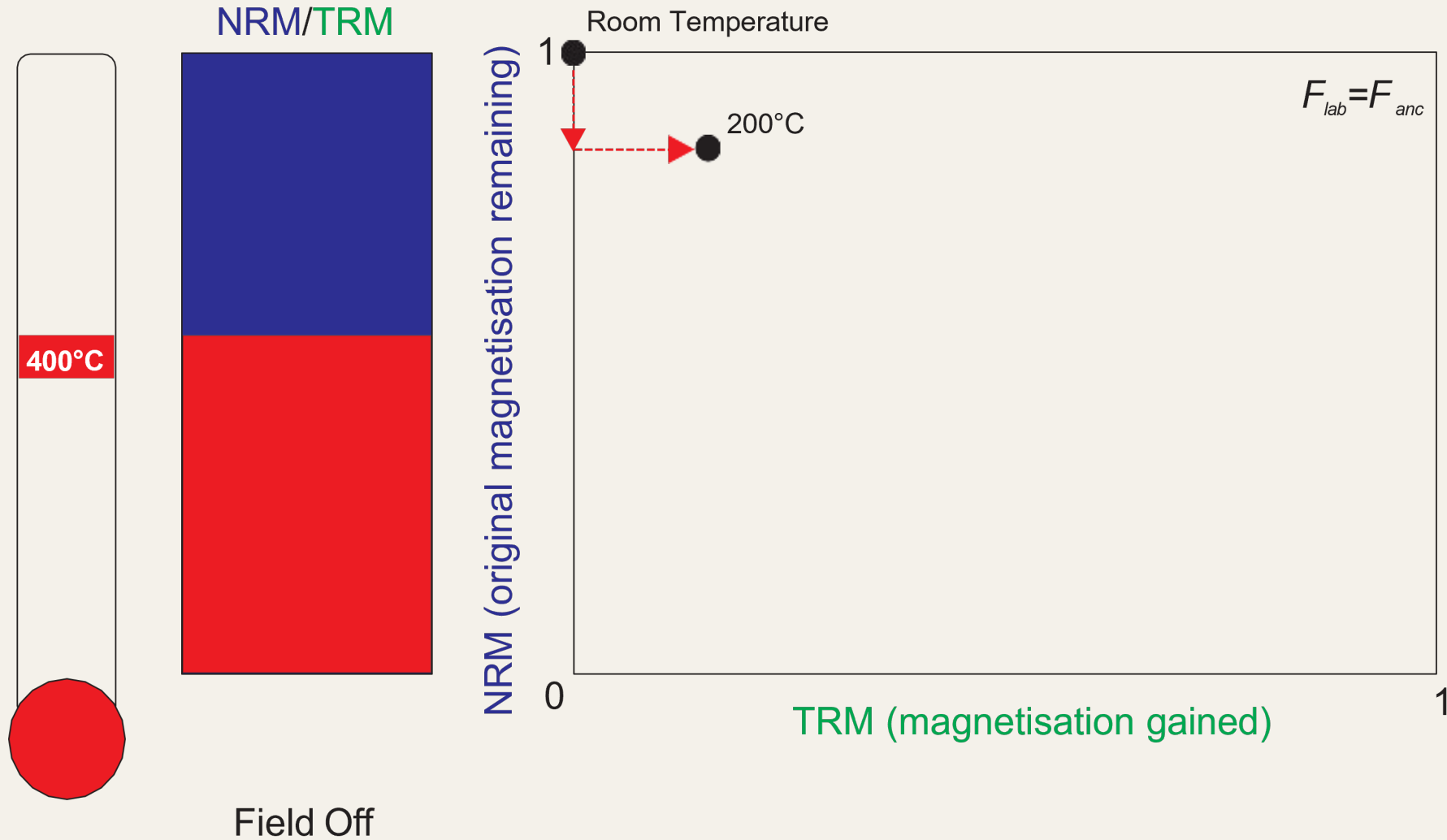
Coe-Thellier Method (on Arai-Nagata plot)



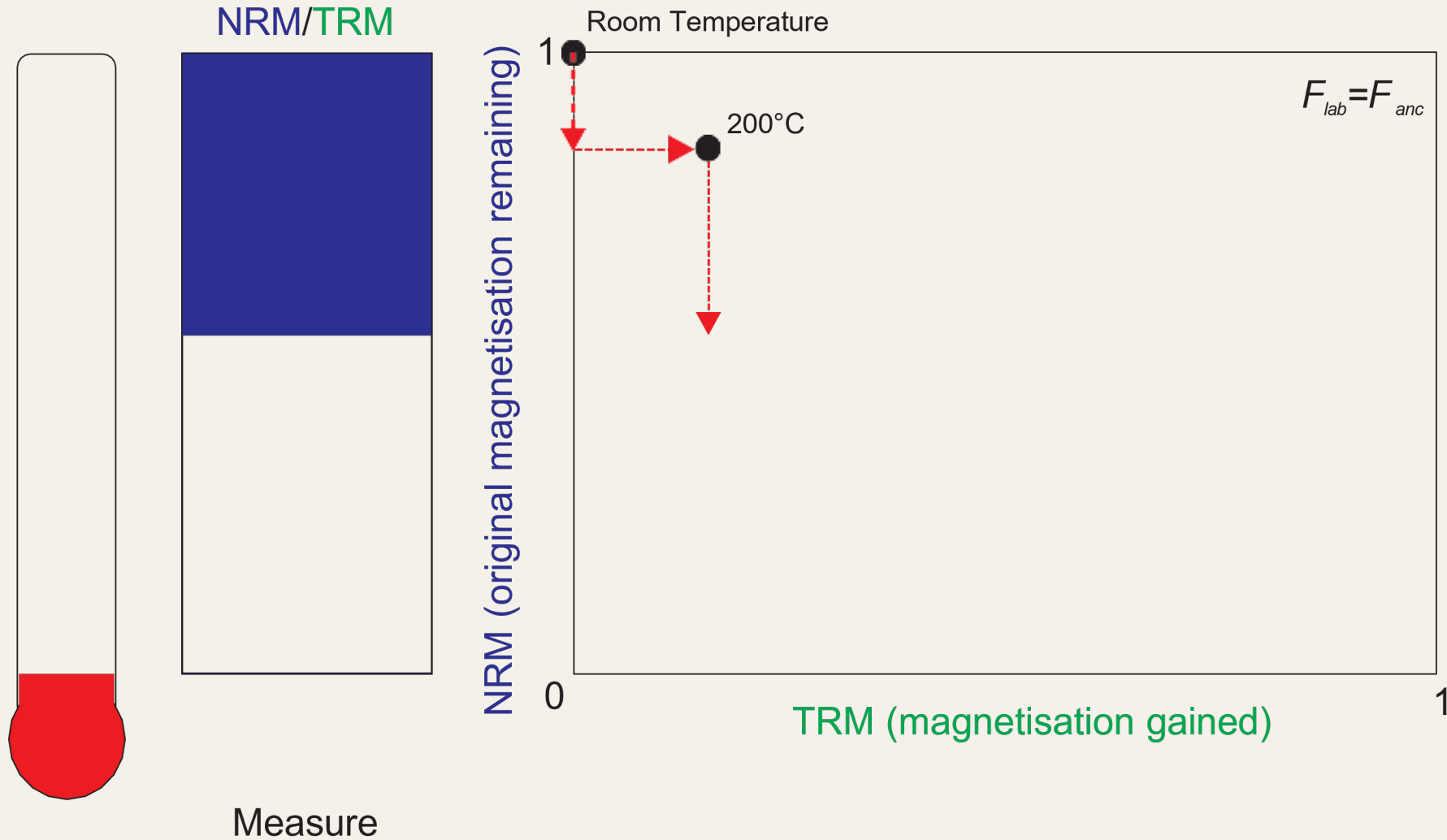
Coe-Thellier Method (on Arai-Nagata plot)



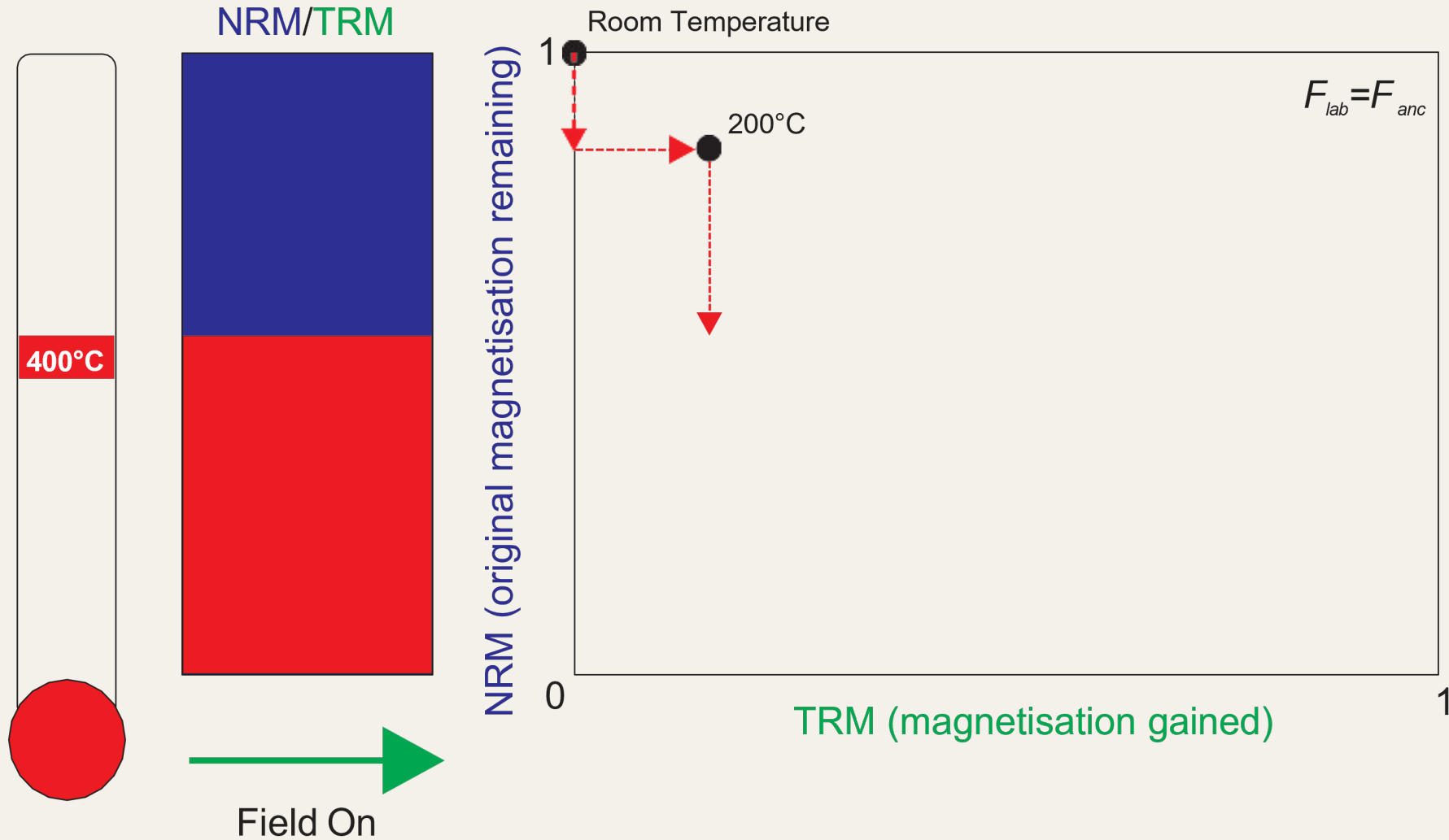
Coe-Thellier Method (on Arai-Nagata plot)



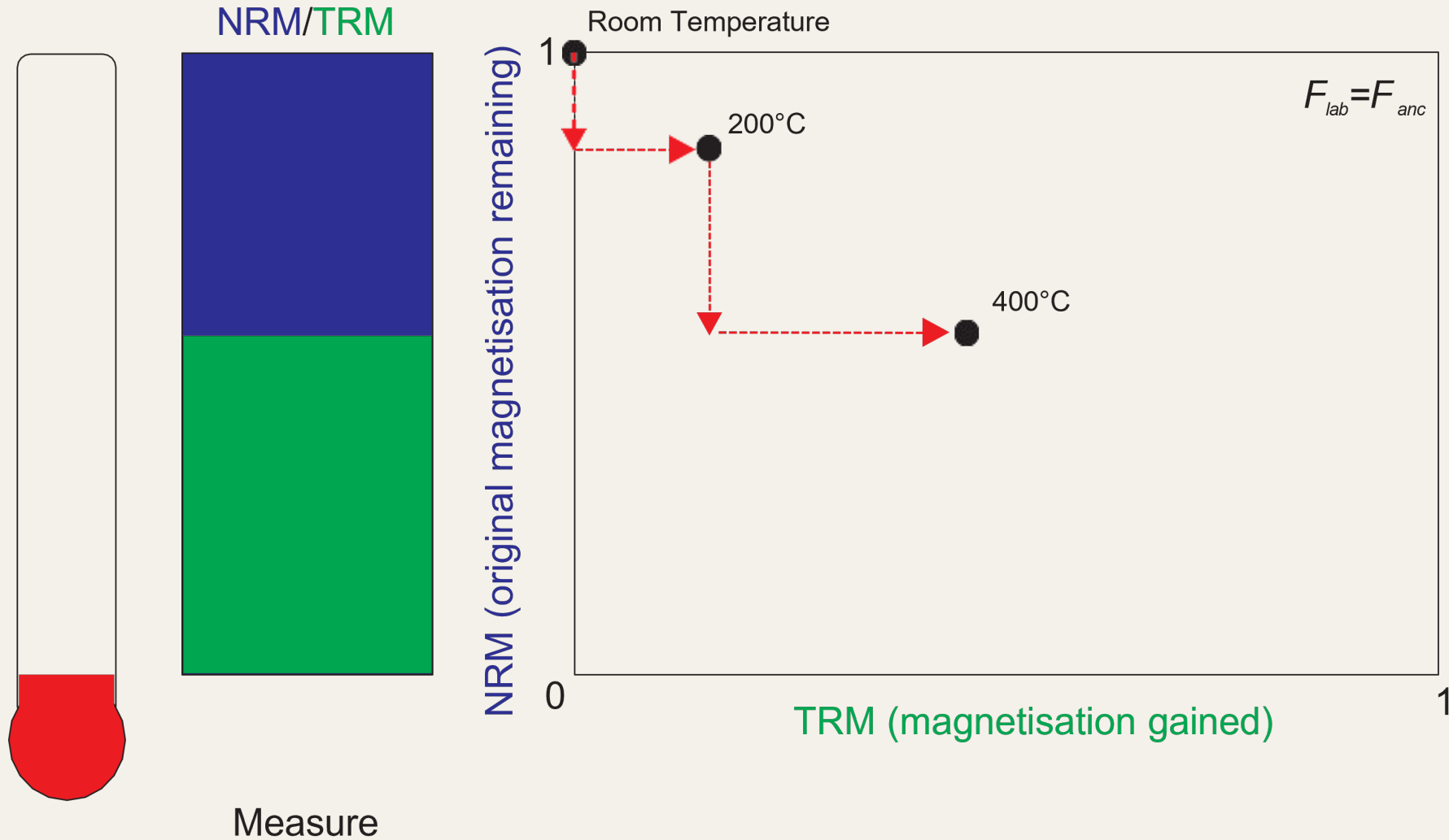
Coe-Thellier Method (on Arai-Nagata plot)



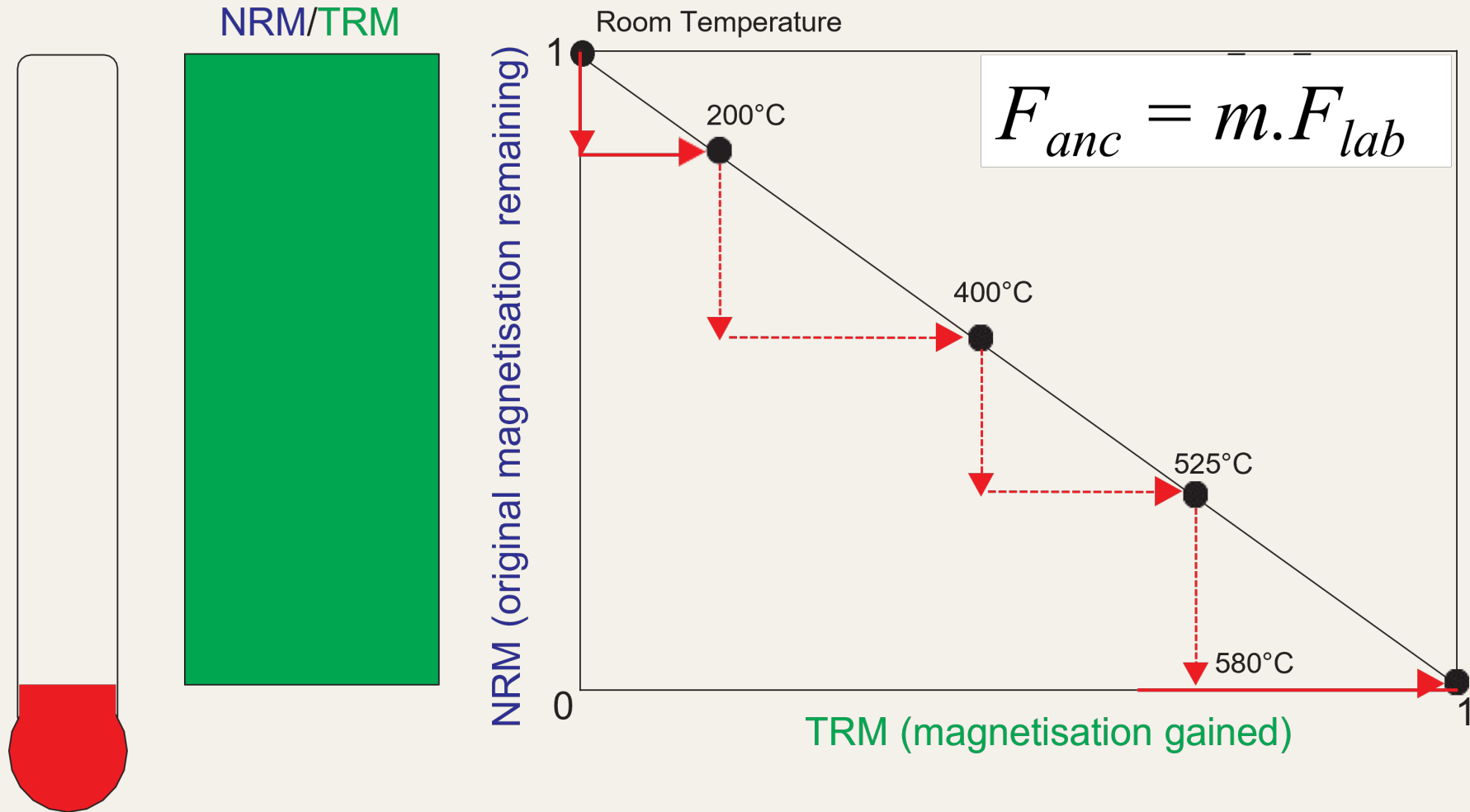
Coe-Thellier Method (on Arai-Nagata plot)



Coe-Thellier Method (on Arai-Nagata plot)



Coe-Thellier Method (on Arai-Nagata plot)



The Thellier Family Variations

II

in field -ve → in field +ve

Thellier-Thellier*

ZI

Zero field → in field

Classical Coe sequence

IZ

In field → zero field

Reversed order

IZZl

Alternating IZ / ZI

Makes pTRM-tail zigzags visible

Protocols themselves creates diagnostic contrast between ideal and nonideal responses.

*hard to create a zero-field environment at the time, so used Earth's magnetic field in opposing directions to the samples

Does it work? Test on synthetic hematite

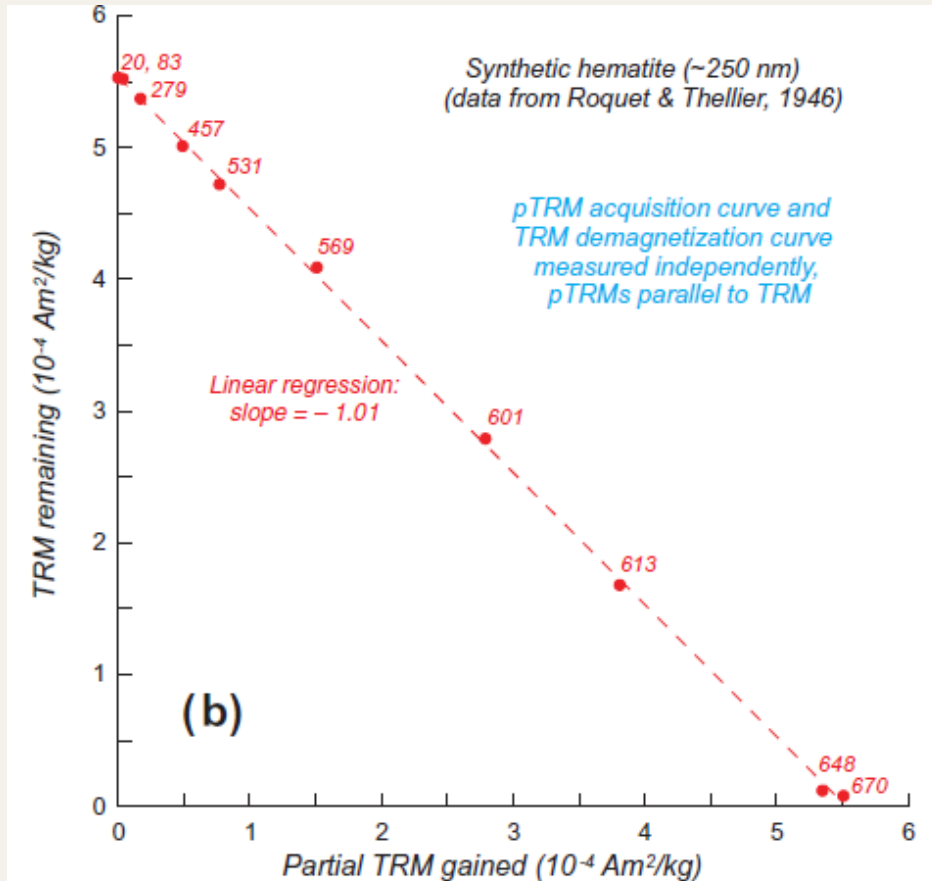


Fig. 5. Roquet and Thellier's (1946) data for synthetic nanoparticle hematite. The TRM demagnetization and pTRM acquisition curves were measured separately, not in interwoven double heatings. Perfect mirror symmetry (inverted TRM curve matching pTRM curve) in (a) and the linear TRM – pTRM graph in (b) confirm SD behavior. The concentration of high T_B 's contrasts with the even distribution of T_B 's in Fig. 4.

Excellent Review Paper:

Dunlop, D., Physical basis of the Thellier–Thellier and related paleointensity methods, *Physics of the Earth and Planetary Interiors* 187 (2011) 118–138

Does it work? Test on a modern brick in known field

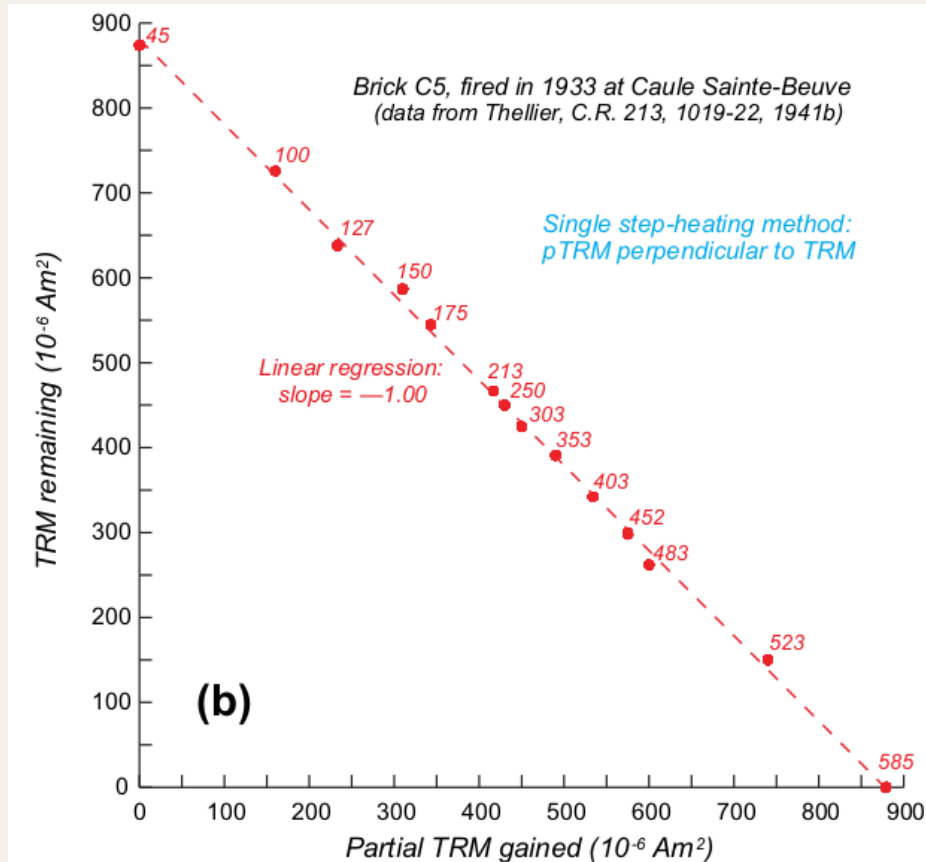


Fig. 4. Thellier's (1941b) single step-heating procedure applied to a modern brick. TRM was produced by the Earth's field in the X-Z plane. The brick was then rotated 90° for step-heatings with the field in the X-Y plane. At each heating step, TRM remaining was measured along Z, pTRM gained was measured along Y, and the X-component of TRM was unchanged. Results are shown in a Roquet plot (moment vs. temperature) in (a) and an Arai plot in (b). SD behavior is evident from the straight-line Arai plot and the mirror-image symmetry of the Roquet plot.

Excellent Review Paper:

Dunlop, D., Physical basis of the Thellier-Thellier and related paleointensity methods, *Physics of the Earth and Planetary Interiors* 187 (2011) 118-138

Does it work? Real almost perfect data

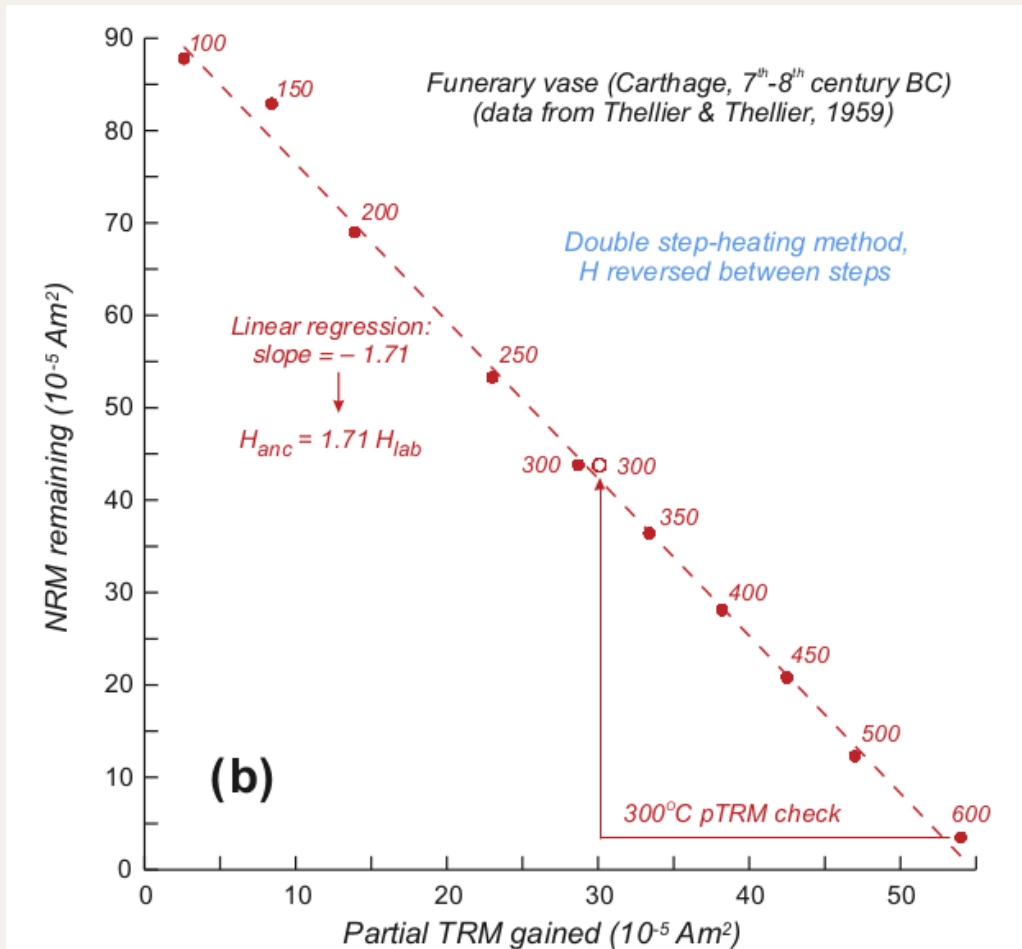


Fig. 7. Thellier–Thellier double-heating of a sacrificial urn from Carthage. Matching Roquet plot curves between 200 and 500 °C gives $H_{anc} = 1.71 H_{lab}$. The Arai plot over the same range yields a similar value. Note the 300 °C pTRM check, showing only minor alteration as a result of heating to 600 °C.

Excellent Review Paper:

Dunlop, D., Physical basis of the Thellier–Thellier and related paleointensity methods, *Physics of the Earth and Planetary Interiors* 187 (2011) 118–138

Thermal alteration changes the recorder during testing

Heating can change mineralogy, oxidation state, grain size, stress state, or magnetic interactions.

Before treatment

Primary carrier population

During experiment

Chemical / physical evolution

What you measure

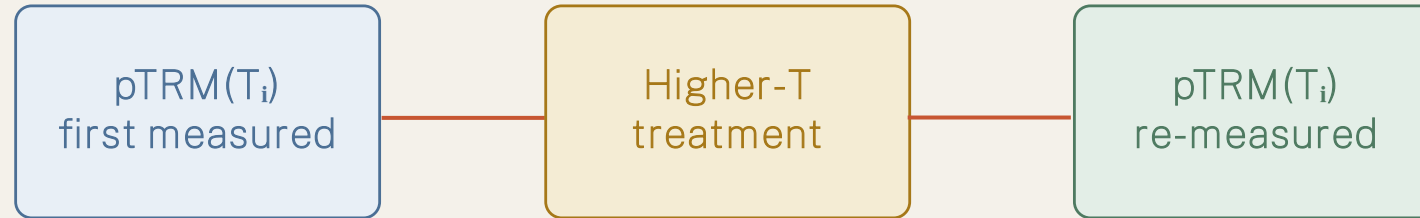
A mixture of original and newly created behaviour

IMPORTANT LIMITATION

A pTRM check can miss alteration whose new unblockers lie above the check temperature.

pTRM checks test alteration

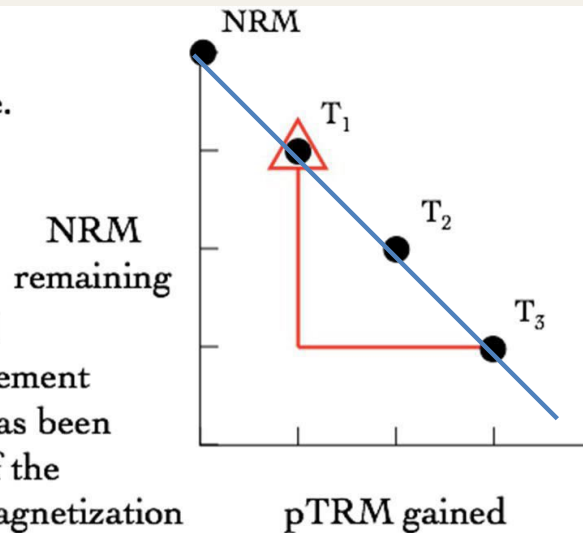
Repeat an earlier in-field step after higher-temperature treatment:



Heat the rock up to T_1 ,
Cool in zero field, measure.

Heat the rock up to T_1 ,
Cool in a known field,

The resulting point should
overlap the earlier measurement
at T_1 . If not, then there has been
a change in the capacity of the
mineralogy to acquire a magnetization



INTERPRET CAUTIOUSLY

A positive check supports repeatability at that temperature. It does not prove that all relevant alteration or domain-state effects are absent.

When Arai plots go Awry

NON-IDEAL BEHAVIOR

This sort of behavior, where the Law of Reciprocity fails results in Arai plots that look like this:

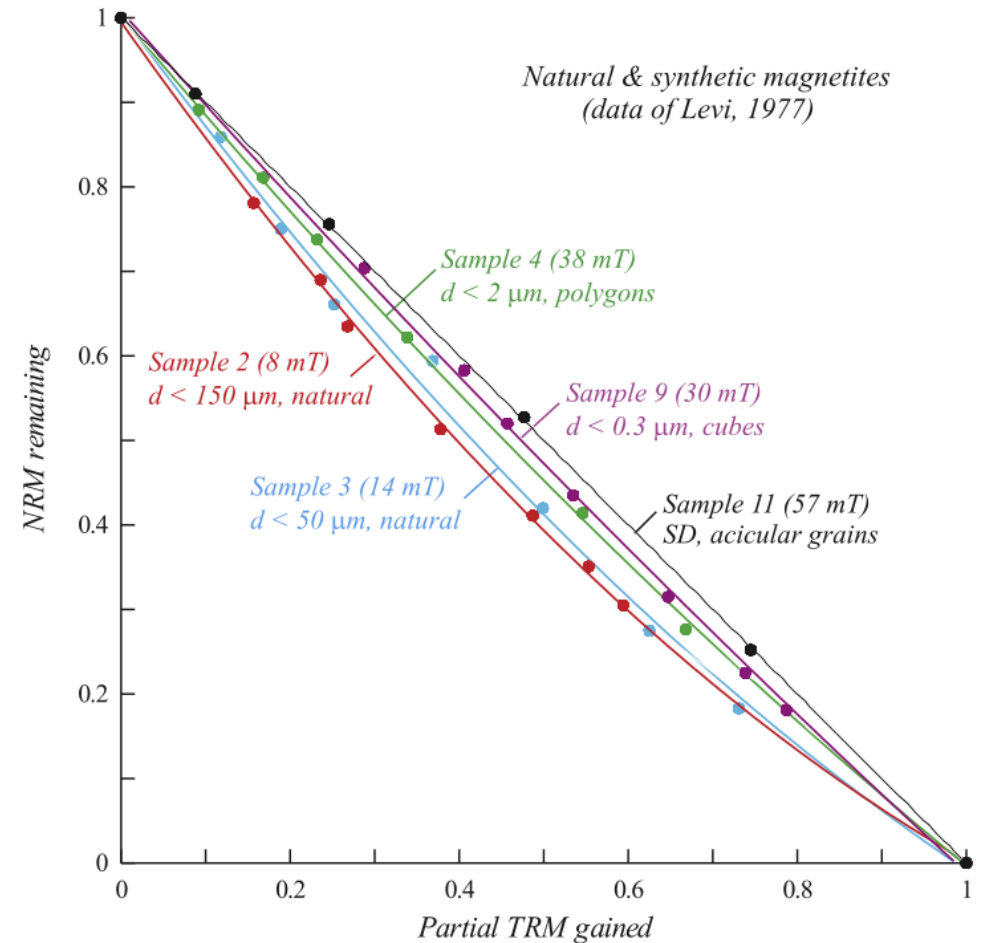
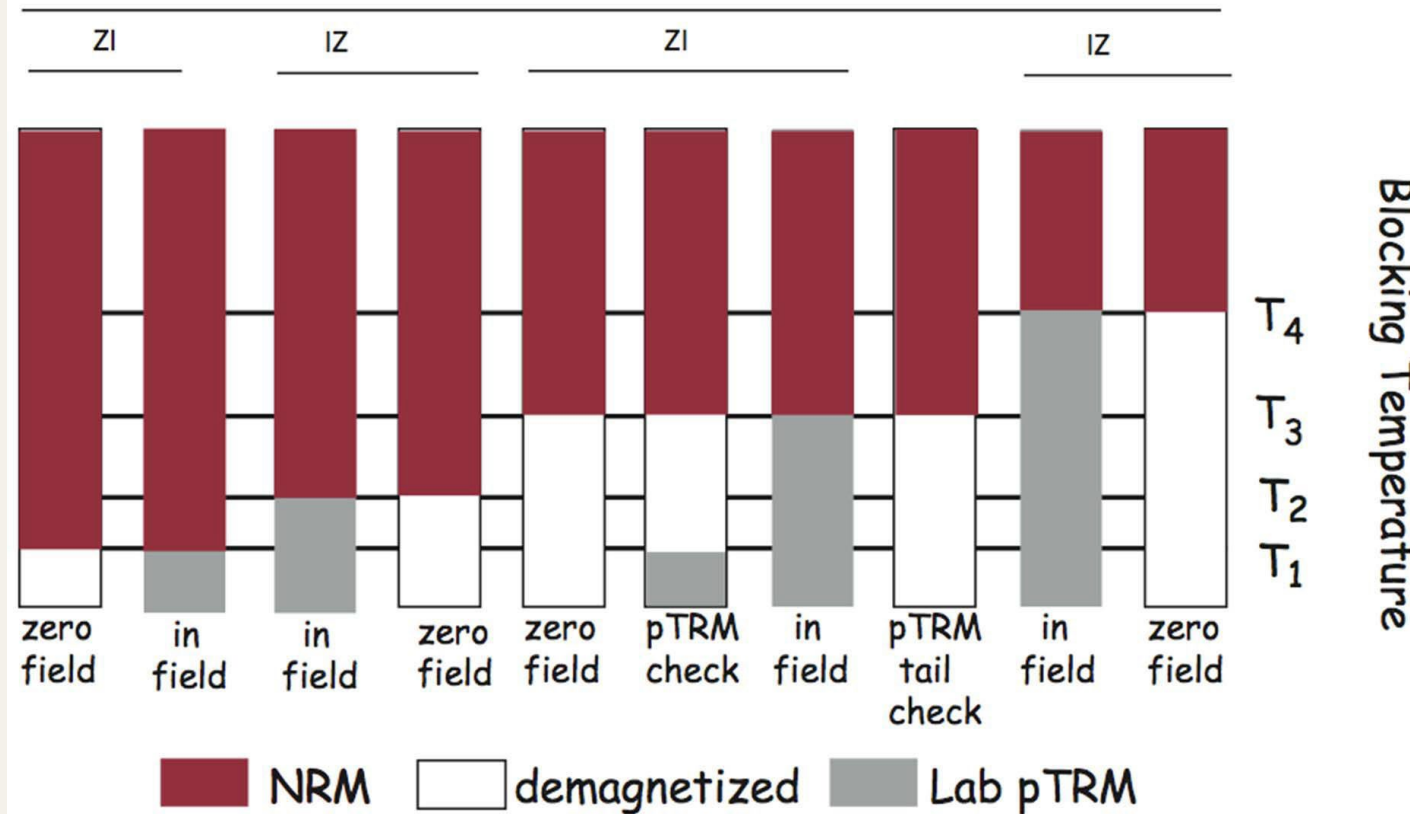


Fig. 13. A selection of Levi's (1977) data showing that the Arai plots of magnetites sag increasingly below the ideal SD straight line as the grain size increases above SD size. The numbers in parentheses are median destructive fields.

This sagging away from the ideal SD line is an expression of pTRM tails (unequal unblockers).

IZZl zigzags and pTRM tails reveal nonideal blocking spectra

Yu, Y., L. Tauxe, and A. Genevey (2004), Toward an optimal geomagnetic field intensity determination technique, *Geochem. Geophys. Geosyst.*, 5, Q02H07, doi:10.1029/2003GC000630.

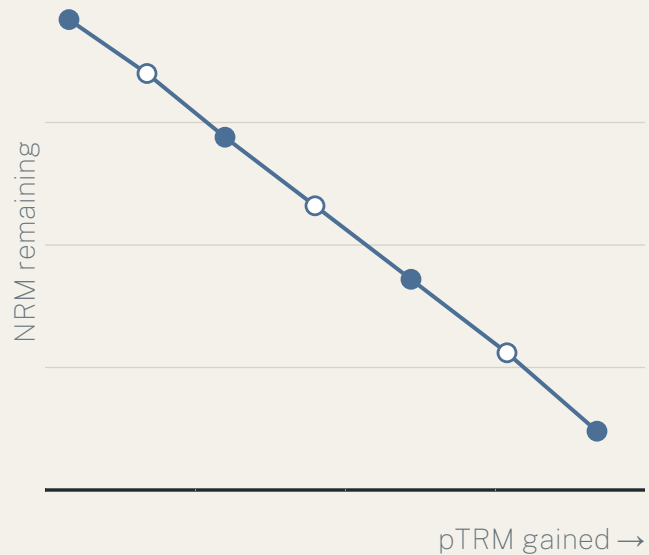


Four heatings (the pTRM check and the pTRM tail check steps) at every other temperature step
Double heatings at the intervening temperature steps

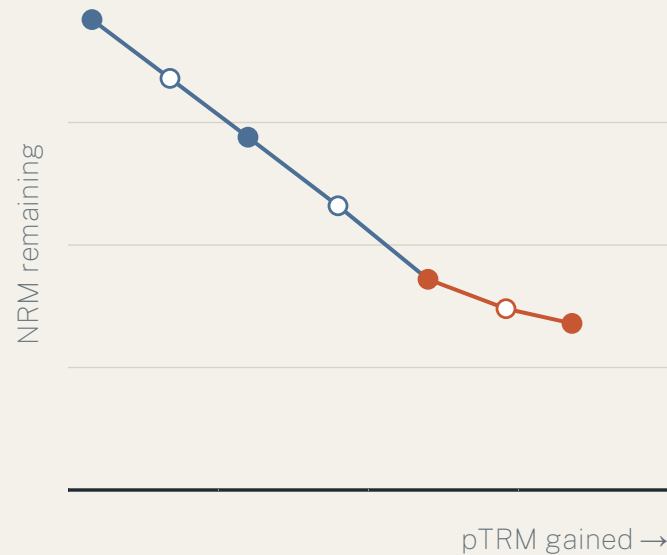
INTERPRETATION

A zigzag is evidence about remanence physics – not an automatic rejection. Use it with mineralogy, checks, and vector behaviour.

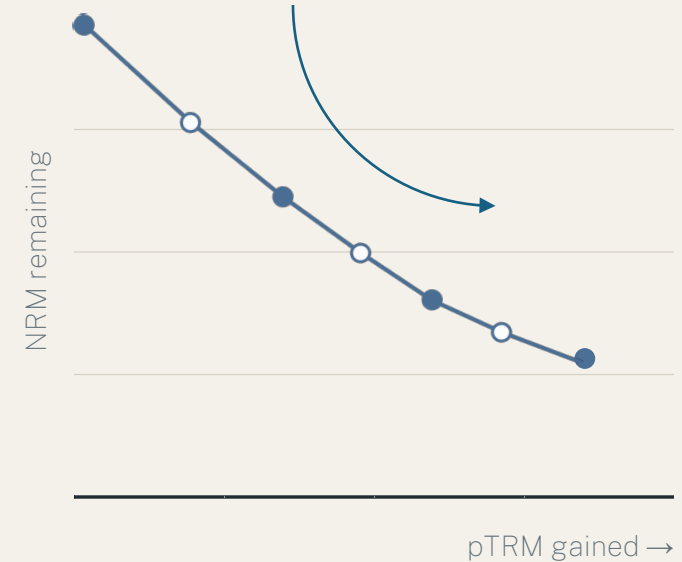
Arai geometry important – linearity underpins the method



linear: promising



late flattening: alteration?



curvature: nonideal domain state?

The correct question is “what mechanism could make this shape?”, followed by an independent test.

Laboratory checks

pTRM check

Has the measured lab remanence changed?

Tail check / IZZI pattern

Does blocking behaviour depart from ideal SD-like reciprocity?

Vector data

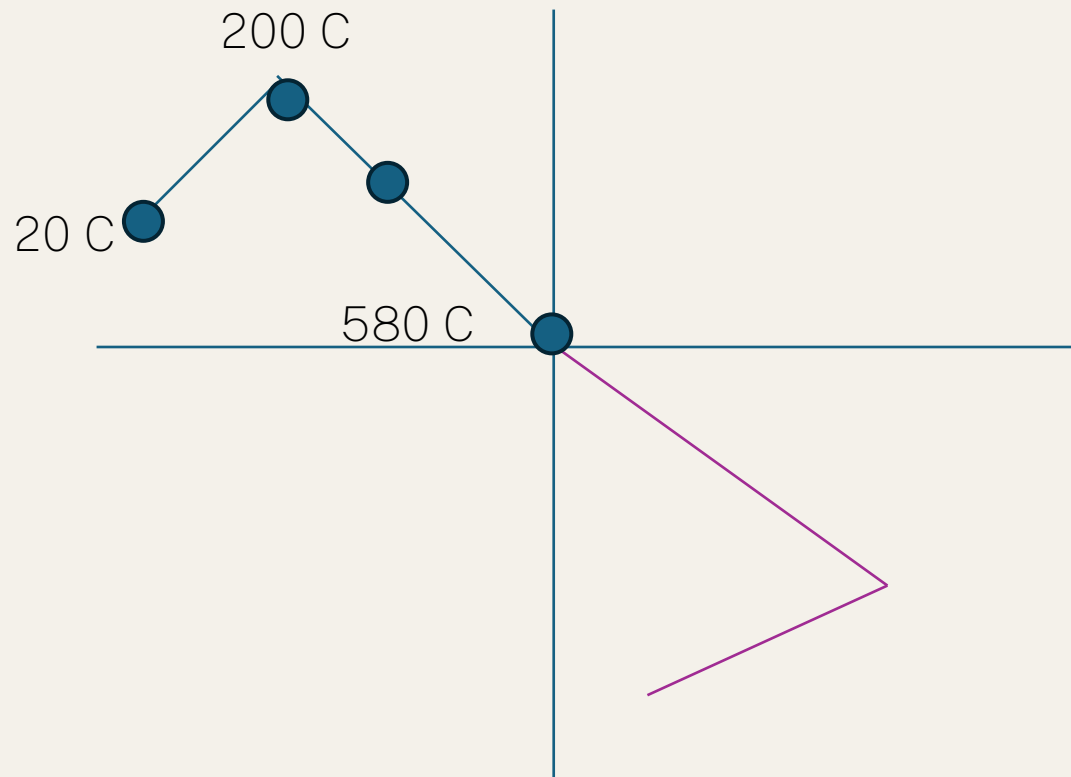
Is the isolated component actually being replaced?

Rock magnetism

Is a plausible stable carrier present?

A result earns confidence through convergent lines of evidence.

Mixed NRM can make the wrong interval look ideal



Two component Zijderveld plot

Component interpretation comes before intensity fitting. Use vector demagnetization data to identify the interval that plausibly carries the primary record.

RED FLAG

An elegant Arai segment is irrelevant if it replaces a secondary component.

Triage before committing to thermal treatment



Screening does not guarantee success; it makes failure modes explicit before expensive heating.

The right decision may be to correct, redesign, or abandon a specimen

Proceed

Primary component, stable mineralogy, coherent checks, plausible Arai interval.

Correct

Evidence supports an anisotropy or cooling-rate correction with uncertainty.

Redesign / reject

Unresolved alteration, mixed remanence, severe nonideal behaviour, or no reproducible interval.

02

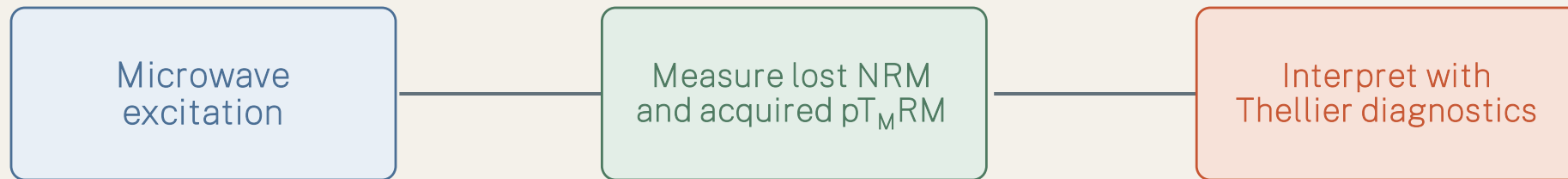
PROTOCOL CHOICES

Method choices

No method is intrinsically “best”; each method trades off alteration, non-ideal behavior sensitivity, and effort.

Microwave Thellier reduces bulk thermal exposure

Microwaves can preferentially excite magnetic material while reducing whole-sample heating; the experimental logic remains Thellier-type.



RECENT SYNTHESIS

Lloyd et al. (2023) review found no site-level statistical bias of microwave data relative to thermal and known-field comparisons, supporting use within a multi-method design.

Shaw methods with paired ARM alteration monitoring



ARM ratios monitor changes caused by thermal treatment; low-temperature demagnetization is often used to reduce multidomain remanence effects.

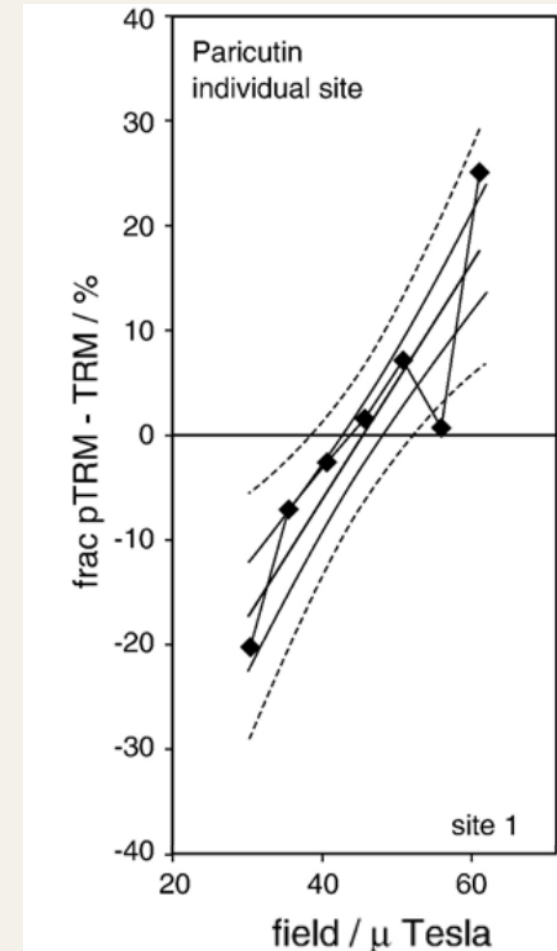
Multispecimen methods – Domain independent?

Instead of repeatedly heating one specimen, heat each sample once and apply different field treatment steps (Dekkers and Böhnel, 2006; Fabian and Leonhardt, 2010).



TRADE-OFF

Less cumulative heating per specimen can help fragile material; the method instead depends strongly on specimen-to-specimen equivalence.



Paleointensity methods: thermal burden vs diagnostic power

Method	Thermal burden	Primary advantage	Central caution
Thermal Thellier / IZZI	High	Strong, direct pTRM checks	Alteration; nonideal domains
Microwave Thellier	Low/Moderate	Less bulk thermal exposure	Microwave-specific calibration / handling
Shaw	High	Efficient alteration monitoring through ARM and LT demag.	Alteration and ARM vs TRM inequality
Multispecimen	Lower per specimen	Reduced cumulative heating	Inter-specimen comparability
Pseudo-Thellier	No thermal treatment	Quick and no alteration	Thermal-AF inequality. Calibration and material dependence

Different experimental strategies

Basaltic lavas

Often thermal or microwave
Thellier; triage mineralogy, grain
size, alteration risk.

Archaeological materials

Usually high-quality TRM;
correct anisotropy and cooling
rate carefully.

Single crystals / silicates

Source-local methods can target
inclusions; low moments and
heterogeneity become central.

Pick the most appropriate method for the material

04

ANALYSIS

Straight lines to testable interpretations

The difficult analytical decision is not simply where to draw a best-fit line; it is understanding what remanence is reliable.

Analysis should test a model of recording behaviour

Observed

Arai plot, vector data, pTRM checks, rock-magnetic context

Hypothesis

Primary NRM is proportionally replaced by a stable laboratory pTRM

Tests

Fit quality, alteration checks, curvature, vector integrity, independent screening

Inference

Field estimate plus its uncertainty, limitations, and exclusions

Paleointensity statistics have complementary jobs

FRAC

How much NRM is represented by the fitted interval?

β / σ

How precise is the fitted slope?

q

A compound quality factor; useful but never sufficient alone.

MAD / DANG

Does vector behaviour support a coherent component?

DRAT / CDRAT

How much do pTRM checks deviate?

Metrics are lenses. Agreement among physical and statistical evidence is stronger than any threshold crossed in isolation.

Criteria families

TTP-style

Flexible, context-rich set of paleointensity statistics.

CCRIT

More restrictive thresholds calibrated against known-field data.

Study-specific

Explicitly justified for the material and tested with independent evidence.

BEST PRACTICE

Declare criteria before examining the preferred site mean; show sensitivity to reasonable alternatives.

Uncertainty accumulates through a hierarchy of decisions

Measurement

instrument noise; repeatability

Specimen

fit interval; curvature; checks

Correction

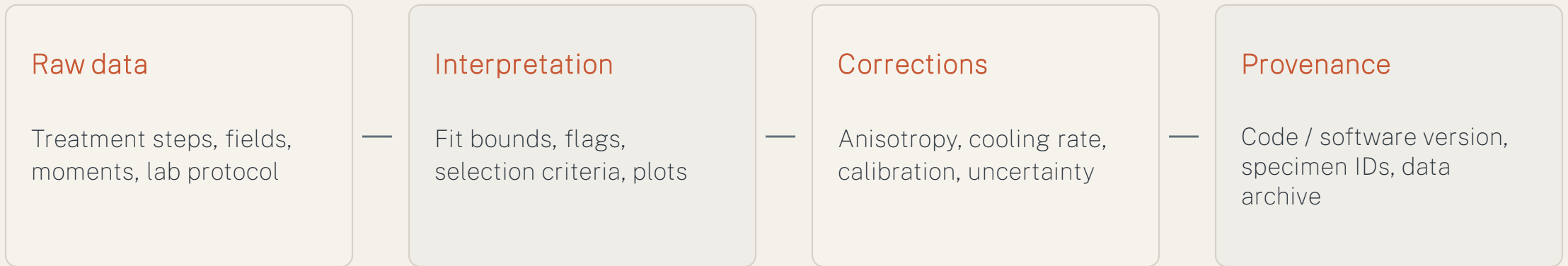
anisotropy; cooling-rate; calibration

Site / age

between-specimen scatter; geological representativeness

A narrow regression confidence interval is not the uncertainty of the palaeofield.

A reproducible workflow records every choice that can move the answer



This is not administrative overhead. It is the experimental record that lets others evaluate your inference.

Modern analysis tools

PmagPy / Thellier GUI

Desktop + notebook-capable analysis of Thellier-type data with standard statistics and MagIC interoperability.

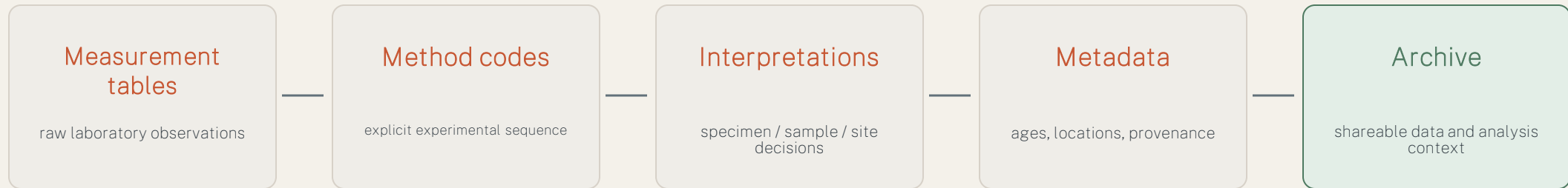
Paleointensity.org

Browser-based import, visualisation, manual and automated specimen interpretations across several methods.

Custom notebooks

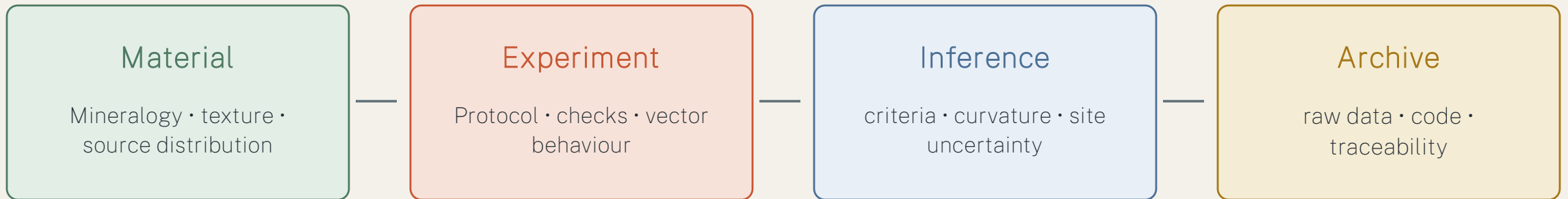
Versionable calculations, transparent figures, sensitivity analyses, and direct links to project metadata.

MagIC-ready records preserve enough context for reuse and reassessment



The point is not only to publish an intensity — it is to publish a testable claim about the ancient field.

Paleointensity needs an evidence chain across scales



A field estimate becomes robust when all four layers point to the same geological story.

Mature, promising, and cautionary methods

Mature practice

Thellier / IZZI diagnostics, correction experiments, provenance-aware analysis.

Useful specialised options

Microwave, Shaw, multispecimen, pseudo-Thellier when their assumptions are tested.

Promising integration

QDM / microscopy-informed selection, curvature-aware site inference, open notebook workflows.

Do not outsource the physics

Automation or machine learning can help triage; they cannot certify a primary stable recorder on their own.

A publication-ready result tells the reader what was measured and why it was believed

Specimen evidence

Arai / vector plots, check outcomes, chosen temperature interval

Corrections

anisotropy, cooling rate, calibration source, propagated uncertainty

Site inference

number of specimens, within-site scatter, treatment of outliers / curvature

Reproducibility

raw data, protocol details, software / code, data repository ID

Three claims to carry into your next paleointensity project

- 1 A straight Arai line is a starting point, but not a conclusion.
- 2 Protocol choice is an explicit decision about anticipated failure modes.
- 3 Modern analysis improves palaeointensity when it preserves evidence, uncertainty, and provenance.

What would make you trust this field estimate?

- 01 Which failure mode is most likely for the material?
- 02 What would the pTRM, vector, and rock-magnetic evidence need to show?
- 03 Which alternative protocol or analysis would genuinely add independent evidence?